TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Horizontal Component of the Earth's Field and Magnetic Moment of a Magnet by Deflection and Vibration Magnetometers

Two instruments, two combinations: the deflection magnetometer yields M/B_h, the vibration magnetometer yields $M{\cdot}B_h$ — and between them, both M and B_h fall out.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 8 · LEVEL 1 TERM 2

1. Aim and theory

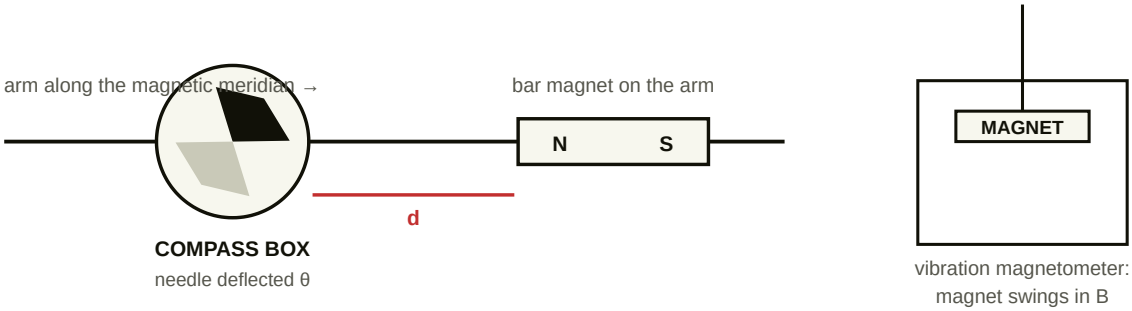
Aim: to determine B_h, the horizontal component of the earth's magnetic field at the laboratory, and M, the magnetic moment of a bar magnet. Neither instrument alone can do it — each measures a combination. In the **deflection magnetometer** (tan-A position), the bar magnet lies on the arm of a compass box at distance d from the needle, its axis along the line to the needle. Its field at the needle, $B = (\mu_0/4\pi)\cdot 2Md/(d^2 - l^2)^2$ (l = half the magnet's length), acts perpendicular to B_h, so the tangent law gives

$$B_h \tan\theta = (\mu_0/4\pi)\cdot 2Md/(d^2 - l^2)^2 \quad \rightarrow \quad M/B_h \text{ from } \theta$$

In the **vibration magnetometer** the same magnet hangs horizontally in a stirrup from a torsion-free thread and oscillates in B_h alone. Its period $T = 2\pi\sqrt{(I/(M\,B_h))}$, with I the magnet's moment of inertia about the suspension axis, gives

$$M{\cdot}B_h = 4\pi^2\, I / T^2$$

Multiplying the two combinations gives M^2, dividing gives B_h^2 — the classic pairing that separates the magnet from the planet.



h alone DEFLECTION GIVES M/B_h · VIBRATION GIVES M·B_h · TOGETHER: M AND B_h

Figure 1 — Left: the deflection magnetometer in tan-A position — the magnet on the graduated arm deflects the compass needle by θ. Right: the vibration magnetometer — the same magnet, suspended, timing its swings.

2. Apparatus and procedure

- **Apparatus:** deflection magnetometer (compass box with graduated arms), vibration magnetometer (suspension frame and stirrup), the bar magnet, stopwatch, balance, scale, slide calipers.
- Measure the magnet: mass m, length 2l and breadth b (for its moment of inertia $I = m((2l)^2 + b^2)/12$ about a vertical axis through its centre).
- **Deflection:** set the arms along the magnetic meridian, needle at zero. Place the magnet at distance d (centre to centre) on one arm, read both ends of the needle; repeat on the other arm and with the magnet reversed (four readings kill zero and placement errors). Average θ.
- **Vibration:** suspend the magnet in the stirrup, set it swinging through a small angle in the horizontal plane, and time 20 oscillations (start counting at a passage through the rest position). Repeat and average T.
- Compute M/B_h and $M \cdot B_h$; combine for M and B_h. Repeat the deflection at a second distance as a check.

3. Worked calculation

Magnet: m = 0.040 kg, 10.0 cm × 1.2 cm → $I = 0.040 \times (0.10^2 + 0.012^2)/12 = 3.38 \times 10^{-5}$ kg·m²; half-length l = 0.050 m. Deflection at d = 0.30 m: mean θ = 36°, tan θ = 0.7265.

$$M/B_h = \tan\theta\,(d^2 - l^2)^2 / ((\mu_0/4\pi)\cdot 2d) = 0.7265 \times (0.0875)^2/(2 \times 10^{-7} \times 0.30) \approx 9.27 \times 10^4$$

Vibration: 20 oscillations in 67 s → T = 3.35 s.

$$M \cdot B_h = 4\pi^2 \times 3.38 \times 10^{-5}/(3.35)^2 \approx 1.19 \times 10^{-4}$$

$$B_h = \sqrt{((M\cdot B_h)/(M/B_h))} \approx 3.6 \times 10^{-5}\ \text{T} \quad \cdot \quad M = \sqrt{((M\cdot B_h)(M/B_h))} \approx 3.3\ \text{A}\cdot\text{m}^2$$

4. Precautions

- The magnetometer bench must be magnetically clean: no other magnets, iron, or current leads within a metre or more; the arms aligned to the meridian before starting.
- In vibration, the suspension thread must be untwisted (let it settle with a dummy weight first) and the swing small and purely horizontal — a conical swing falsifies T.
- Distances d are measured centre-to-centre of magnet and needle, on both arms, and the magnet is reversed end-for-end between readings.
- Time many oscillations and start/stop on a crisp transit of the rest position.

Practice (viva preparation)

1. Why can neither magnetometer alone give M or B_h?
2. If the same magnet's period in the vibration instrument is 2.0 s and $I = 3.38 \times 10^{-5}$ kg·m², what is $M\cdot B_h$?
3. Why is the deflection magnetometer used in the tan-A position here rather than tan-B?
4. A stray steel stool is parked near the bench mid-experiment. Which measurement is corrupted, and in what sense?
5. Two identical magnets are strapped together (like poles together) in the vibration stirrup. What happens to T, roughly?

Answer key

1. Deflection involves the magnet's field ($\propto$ M) competing with B_h — only the ratio M/B_h appears. Vibration involves the torque $M\cdot B_h \sin\theta$ — only the product appears. Two equations, two unknowns: the pair separates them.
2. $M\cdot B_h = 4\pi^2 \times 3.38 \times 10^{-5}/4.0 \approx 3.34 \times 10^{-4}$ (units: A·m²·T).
3. In tan-A the magnet's field at the needle is axial (the stronger, $2M/d^3$ form) and perpendicular to B_h, giving larger, cleaner deflections; tan-B uses the weaker equatorial field (M/d^3) along the arm's perpendicular. (Names: tan-A magnet on the arm in line, tan-B magnet on the perpendicular bisector.)
4. Both, potentially: the stool distorts the local field, so the needle no longer responds to B_h plus the magnet alone (deflection readings shift) and the suspended magnet oscillates in a field that is no longer the earth's (T shifts). The whole method presumes the only field present is B_h.

5. Moment doubles (2M), inertia roughly doubles (2I), so $T = 2\pi\sqrt{(2I/(2M \cdot B_h))}$ is essentially unchanged — a neat check that the pair really measures the combination.
